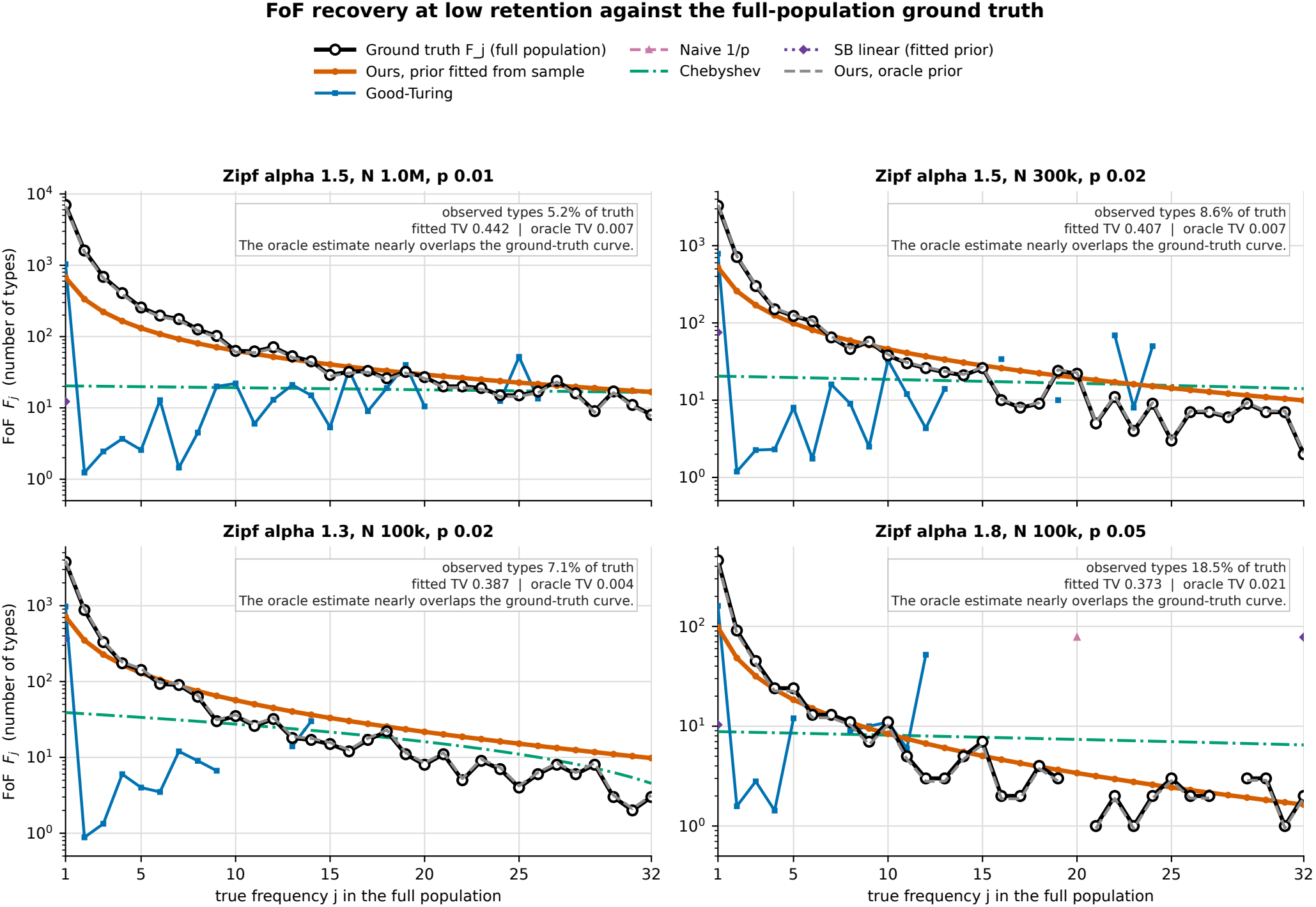
FoF recovery at low retention against the full-population ground truth
Ground truth F_j (full population)
Ours, prior fitted from sample
Good-Turing
Naive 1/p
Chebyshev
Ours, oracle prior
SB linear (fitted prior)
Zipf alpha 1.5, N 1.0M, p 0.01
observed types 5.2% of truth
fitted TV 0.442 | oracle TV 0.007
The oracle estimate nearly overlaps the ground-truth curve.
Zipf alpha 1.5, N 300k, p 0.02
observed types 8.6% of truth
fitted TV 0.407 | oracle TV 0.007
The oracle estimate nearly overlaps the ground-truth curve.
Zipf alpha 1.3, N 100k, p 0.02
observed types 7.1% of truth
fitted TV 0.387 | oracle TV 0.004
The oracle estimate nearly overlaps the ground-truth curve.
Zipf alpha 1.8, N 100k, p 0.05
observed types 18.5% of truth
fitted TV 0.373 | oracle TV 0.021
The oracle estimate nearly overlaps the ground-truth curve.
FoF F_j (number of types)
true frequency j in the full population